

Aasheesh Bajpai
HALL XI, IIT Kanpur,
Kanpur (U.P., India) -208016
Email-id : aasheesh@iitk.ac.in
Email-id : aasheeshbajpai@gmail.com
Mobile No.: 8348323133

Current Position

- **Prime Minister Research Fellow at Aerospace Engineering Department, Indian Institute of Technology Kanpur**
- **Affiliated with Non-Equilibrium Flow Simulation Laboratory, Aerospace Engineering Department, IIT Kanpur**

ACADEMIC DETAILS

Degree/examination	University/Board	Institute	Year	CPI/%
Ph.D Specialization:	<i>Aerospace Engineering</i>			
Ph.D (Course work)	IIT Kanpur	IIT Kanpur	2020	10
Post Graduate Specialization:	<i>Cryogenic Engineering</i>			
Post Graduation	IIT Kharagpur	IIT Kharagpur	2019	8.51
UnderGraduate Specialization:	<i>Mechanical Engineering</i>			
Graduation	AKTU, Lucknow	AKGEC, Ghaziabad	2016	76.1
Intermediate	U.P. Board	JNSVM Bareilly	2011	85.80
High School	U.P. Board	JNSVM Bareilly	2009	85.50

PROJECTS

- **Dusty Gas Flow Dynamics** (Ph.D project)
(Advisor- Prof. Rakesh Kumar Mathpal, March'20-present)
 - Objective :Computational modeling of dusty gas flows by developing an in-house particle based solver. The proposed solver will be developed by combining Discrete Element Method (DEM) code with the in-house Direct Simulation Monte-Carlo (DSMC) code, Non-equilibrium Flow Simulation (NFS) solver..
- **Development of test rig towards experimental investigation of Reverse Brayton Cryocooler used for cooling of High Temperature Superconductors** (M.Tech Project)
(Guide:Prof. Parthasarathi Ghosh , July'18 - May'19)
 - Objective :The objective of this work is to build a laboratory scale reverse Brayton cryocooler to generate performance data at different operating conditions.
- **Design, Fabrication and Review of GO-KART** (B.Tech Project)
(Guide: Asst. Prof. Ankur Tyagi , July'15 - May'16)
 - Objective :The objective of this work is to design, fabrication of GO-KART and improvising by innovative approach
 - Responsibility : Managed a team of 10 engineers.

LIST OF PAPERS IN JOURNALS/conferences

- **Aasheesh Bajpai**, Rohan Dutta, Parthasarathi Ghosh, Parameter Estimation of Equipment for Development of an Experimental Setup of a Reverse Brayton Cryocooler for Cooling High Temperature Superconducting Cables, Presented at National Symposium in Cryogenics and Superconductivity-27 at IIT Bombay, Mumbai, India on 16th to 18th January, 2019, Indian Journal of Cryogenics.
- **Aasheesh Bajpai**, Aman Dhillon, Rohan Dutta, Parthasarathi Ghosh, Root cause analysis of early performance deterioration of an existing helium liquefier using process simulation, Full paper accepted for presentation at 15th Cryogenics 2019, IIR international conference, April 7-11, 2019, Prague, Czech Republic, Europe.

- Aman Dhillon, **Aasheesh Bajpai**, Parthasarathi Ghosh, The effect of HTS heat rejection conditions on performance of reverse Brayton cryocooler Full paper accepted for poster presentation at 15th Cryogenics 2019, IIR international conference, April 7-11, 2019, Prague, Czech Republic, Europe.

TRAINING

- **In workshop training Indian Railways workshop, Izzatnagar, Bareilly(June-July 2014)**
 - Braking system in locomotive (Air brakes, pneumatic brakes), Working of engine of locomotive, carriage maintenance.
- **Vocational Training in NTPC, Dadri (July-August 2015)**
 - Boiler maintenance and Ash handling department, Learnt about Power plant process, Rankine cycle and various equipment used etc. Also submitted a project report on Cooling Tower.

TECHNICAL SKILLS AND EXPERTISE

- **Languages** (C, C++), **Softwares** (ANSYS fluent, ANSYS CFX, Aspen- HYSYS, Aspen-EDR, HYSYS- Dynamics, Solid-works, Pro-E, EES) **Tools** (MS Office) **Expertise subject** (Thermal engineering, Cryogenic turboexpander, CFD, Hydrogen technology, Process Plant and Simulation)

CERTIFICATIONS

- Automation technologies-Basic level (January-2015) (Rexroth Bosch Group)
 - Topics-Industrial Hydraulics, Industrial Pneumatics, Programmable Logic controllers
- Training program on Pro/Engineer (RICC Akgec)[Feb 2014 to Feb 2015]

EXTRACURRICULAR ACTIVITIES

- Presented a oral presentation in the 15th CRYOGENICS 2019 IIR International Conference exhibition at Prague, Czech Republic.
- Presented a oral presentation in the NSCS-27 hosted by IIT Bombay.
- Participated and won robot racing car arena race.
- Active member of ISTE 2014-2016.
- Participated and won first runner-up title in IGC LPU and managed a team of 10 engineering students.
- Worked as a teaching assistant under Prof. Parthasarathi Ghosh, IIT Kharagpur